Strategic Behavior

Final Exam • Spring 2024

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3 hours • 100 points

Name: _____ Section / GSI: _____

Read this first.

- This exam has five questions and is cumulative. Budget roughly 30–35 minutes per question.
- You may use two double-sided pages of handwritten notes. A basic (non-graphing) calculator is permitted.
- Explain your reasoning. State clearly what game you are analyzing and which solution concept you use.
- Players are risk neutral. "Equilibrium" in a dynamic game means subgame-perfect equilibrium unless stated otherwise.

Question 1: The Tractor Auction (20 points)

A farm is selling a vintage tractor. Four collectors are interested. Each collector's value (in thousands of dollars) is private and drawn independently from the uniform distribution on $[0, 60]$.

(a) *(4 pts)* In a sealed-bid second-price auction, show that bidding one's true value is a weakly dominant strategy.

(b) *(6 pts)* In a sealed-bid first-price auction, show that it is a symmetric Bayesian Nash equilibrium for each collector to bid $b(v) = \frac{3}{4}v$. (You may restrict attention to deviations to bids in $[0, 45]$; explain briefly why bids above 45 cannot be optimal.)

(c) *(5 pts)* Compute the seller's expected revenue in each auction.
Hint: for n independent draws from $U[0,1]$, the expected value of the k-th highest draw is $\frac{n+1-k}{n+1}$.

(d) *(5 pts)* The seller considers a second-price auction with a reserve price of $r = 30$: the tractor is not sold if every bid is below 30; if exactly one bid is at least 30 the winner pays 30; otherwise the winner pays the second-highest bid. Assuming truthful bidding, compute the seller's expected revenue.
Hint: the density of the second-highest of four independent $U[0,1]$ draws is $12x^2(1-x)$ on $[0,1]$.

Question 2: Deterring Entry into the Ready-Mix Market (18 points)

An incumbent concrete supplier (firm I) faces a potential entrant (firm E). Inverse demand is $P = 90 - Q$ and both firms have zero marginal cost. The timing is:

1. Firm I chooses its output $q_I \geq 0$, which is observed by firm E.
2. Firm E decides whether to enter. Entry costs a sunk fee $F \geq 0$. If it enters, it chooses $q_E \geq 0$; the price is then $90 - q_I - q_E$. If it stays out, firm I is alone in the market.

Assume firm E stays out when it is indifferent between entering and staying out.

(a) *(4 pts)* Given q_I, find firm E's optimal output if it enters and its profit from entering (net of F).

(b) *(5 pts)* Suppose $F = 0$, so that E always enters. Find the SPE outputs, the price, and both firms' profits.

(c) *(5 pts)* Suppose $F = 225$. Find the smallest q_I that deters entry. Should firm I deter entry or accommodate it? Describe the SPE outcome.

(d) *(4 pts)* Repeat part (c) for $F = 25$.

Question 3: Can the Cement Cartel Hold? (20 points)

Firms in a cement market face inverse demand $P = 130 - Q$ and each has constant marginal cost 10. Every year they choose quantities simultaneously; the stage game is repeated infinitely often, and firms discount future profits by $\delta \in (0, 1)$ per year.

(a) *(4 pts)* With two firms, find the stage-game (Cournot) Nash equilibrium quantities and profits.

(b) *(3 pts)* The two firms would like to split the monopoly output equally. Find each firm's quantity and profit under this collusive agreement.

(c) *(7 pts)* Each firm uses a grim-trigger strategy: produce the collusive quantity as long as both have always done so; otherwise produce the Cournot quantity forever. Find the optimal one-period deviation from collusion, its profit, and the smallest δ for which collusion is sustained in an SPE.

(d) *(6 pts)* A third, identical firm enters. Repeat parts (a)–(c) with three firms (collusion now splits the monopoly output three ways). Does entry make collusion easier or harder to sustain?

Question 4: Funding the Community Garden (22 points)

Three neighbors can contribute money to a shared community garden. Neighbor i contributes $g_i \geq 0$, and the garden's quality depends on the total $G = g_1 + g_2 + g_3$. Neighbor i's payoff is

$$u_i = w - g_i + a_i \ln G,$$

where $w = 100$ is each neighbor's budget (large enough never to bind). The neighbors care about the garden to different degrees: $a_1 = 30$, $a_2 = 20$, $a_3 = 10$. Contributions are chosen simultaneously. (If $G = 0$, the payoff is $-\infty$, so in any equilibrium $G > 0$.)

(a) *(5 pts)* Let G_{-i} denote the total contribution of the other two neighbors. Derive neighbor i's best response $g_i(G_{-i})$.

(b) *(6 pts)* Find all Nash equilibria. Who contributes, and how large is the garden budget G?

(c) *(5 pts)* Find the budget G^{**} that maximizes the sum of the three payoffs. Compare it with the equilibrium G and explain the difference.

(d) *(6 pts)* Now suppose $a_1 = a_2 = 20$ and $a_3 = 10$. Find all Nash equilibria. Does the total G differ across them?

Question 5: The Transit Contract (20 points)

The city of Port Ellison (C) and the transit workers' union (U) bargain over how to divide a budget surplus of \$12 million. Offers alternate: the city proposes a split in period 1; if the union rejects, the union proposes in period 2; and so on. A proposal is either accepted, ending the game, or rejected. Each party discounts payoffs by its own factor per period: $\delta_C = 0.8$ for the city and $\delta_U = 0.5$ for the union (whose members lose wages during a dispute). Assume a party accepts whenever it is indifferent.

(a) *(8 pts)* Suppose bargaining can last at most three periods (city, union, city); if the city's period-3 offer is rejected, both get 0. Find the SPE by backward induction. What split is proposed in period 1, and is it accepted?

(b) *(8 pts)* Now suppose there is no deadline. Find the stationary SPE: let x be the city's share when it proposes and y the city's share when the union proposes. Derive x and y and give the period-1 split in millions of dollars.

(c) *(4 pts)* The union builds a strike fund that raises its discount factor to $\delta_U = 0.75$. Recompute the infinite-horizon split. Explain in one sentence why the union gains.

End of exam. Have a good summer.